

HETTINGER, ND, USA

WMO: 727584

Lat: 46.014N

Lon: 102.655W

Elev: 2705

StdP: 13.32

Time Zone: -7.00 (NAM)

Period: 99-19

WBAN: 94038

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-15.4	-9.5	-21.0	1.9	-14.3	-15.8	2.6	-9.3	34.3	33.2	31.4	28.7	7.3	300	0.711

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	28.3	94.3	67.1	90.5	66.4	87.2	65.5	71.8	85.9	69.7	83.5	68.0	81.7	12.2	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.5	112.0	78.9	65.2	103.1	76.4	63.3	96.4	74.0	37.3	85.9	35.4	83.3	33.8	81.6	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 31.6	27.0	24.1	DB	-26.5	100.8	7.0	4.8	-31.5	104.3	-35.6	107.1	-39.5	109.8	-44.6	113.3	(4)
(5)			WB	-25.9	75.7	5.5	3.1	-29.9	77.9	-33.1	79.7	-36.2	81.4	-40.2	83.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.9	17.9	17.8	30.4	41.9	52.1	62.3	70.7	67.9	58.4	43.7	31.2	19.2		
	DBStd	21.53	13.90	14.10	13.15	9.95	8.51	7.15	6.47	6.96	8.93	9.73	11.66	13.15		
	HDD50	4648	996	900	612	274	76	3	0	0	24	241	566	956		
	HDD65	8487	1461	1320	1072	694	406	132	20	50	236	662	1013	1421		
	CDD50	2070	0	0	5	31	142	372	642	556	275	44	3	0		
	CDD65	436	0	0	0	1	7	51	197	142	37	1	0	0		
	CDH74	6728	0	0	4	36	177	821	2702	2103	811	71	3	0		
	CDH80	2884	0	0	0	7	37	271	1268	954	330	17	0	0		
	WSAvg	10.3	10.7	9.8	11.0	11.8	11.8	10.3	9.2	9.1	9.7	10.4	9.9	10.2		
	PrecAvg	15.8	0.3	0.3	0.6	1.6	2.8	3.4	2.1	1.5	1.4	1.1	0.4	0.3		
	PrecMax	27.0	1.0	1.5	2.1	5.3	8.2	8.0	4.9	3.7	6.1	5.0	1.6	0.9		
(15) Precipitation	PrecMin	8.2	0.0	0.0	0.1	0.0	0.2	0.7	0.1	0.1	0.1	0.0	0.0	0.0		
	PrecStd	4.1	0.2	0.3	0.4	1.2	1.8	1.7	1.3	1.1	1.4	1.3	0.4	0.2		
	0.4%	DB	53.6	57.2	72.0	79.2	84.2	93.0	101.7	98.4	94.5	81.7	70.2	53.0		
	MCWB	41.5	44.2	50.2	55.2	63.0	69.0	68.6	67.6	64.5	57.9	50.8	41.4			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2%	DB	44.1	47.1	63.6	71.8	79.1	87.3	95.4	93.2	88.2	73.3	61.4	45.7		
	MCWB	36.8	38.0	47.2	52.9	60.7	66.3	67.7	66.6	62.4	54.0	46.2	37.1			
	5%	DB	39.1	40.9	56.8	65.8	74.1	82.1	91.1	89.8	83.5	66.5	54.8	40.8		
	MCWB	33.9	35.2	44.0	49.7	57.9	64.9	67.5	66.3	60.9	50.9	42.8	34.4			
	10%	DB	36.3	36.6	50.0	60.9	70.0	78.7	87.5	85.7	78.1	61.2	48.8	36.5		
	MCWB	32.3	32.4	40.5	46.9	56.1	62.9	66.7	64.9	59.4	48.6	39.3	31.9			
	0.4%	WB	41.5	44.7	52.2	57.3	66.3	73.4	75.1	73.1	69.1	59.8	50.7	42.0		
	MCDB	51.5	56.4	67.9	74.8	79.1	85.8	89.8	90.0	85.7	76.7	68.4	51.3			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	37.2	39.0	48.0	53.2	62.6	69.7	72.5	70.1	65.9	55.6	47.1	37.8		
	MCDB	43.2	45.7	62.1	68.4	74.7	81.4	86.6	85.5	81.3	69.1	60.5	45.0			
	5%	WB	34.4	35.4	44.5	50.7	59.5	67.3	70.5	68.3	63.2	52.4	43.3	34.8		
	MCDB	38.7	40.7	56.4	63.9	71.6	78.8	84.3	83.0	77.9	64.5	54.1	40.4			
	10%	WB	32.3	32.7	40.9	48.0	57.0	64.8	68.8	66.5	60.5	49.4	39.9	32.2		
	MCDB	36.1	36.7	49.9	59.3	67.8	75.8	82.4	80.9	74.4	60.4	48.1	36.5			
	MDBR	20.7	21.1	22.8	25.3	25.0	25.5	28.3	29.4	29.2	25.6	24.5	20.5			
	5% DB	MCDBR	23.7	26.4	34.7	35.4	32.4	31.6	34.4	35.4	38.4	36.7	34.4	25.6		
(36) Mean Daily Temperature Range	MCWBR	17.6	18.5	20.2	19.6	17.1	15.1	13.0	14.3	16.2	19.7	20.9	18.3			
	5% WB	MCDBR	22.8	24.8	33.4	32.6	28.5	27.5	28.7	30.3	32.6	33.4	33.8	24.7		
	MCWBR	17.5	17.9	20.0	18.6	15.9	14.7	13.5	13.7	14.9	18.2	20.8	18.1			
	taub	0.253	0.259	0.278	0.316	0.338	0.349	0.361	0.366	0.326	0.296	0.264	0.248			
(40) Clear-Sky Solar Irradiance	taud	2.521	2.530	2.530	2.425	2.430	2.433	2.434	2.390	2.485	2.543	2.529	2.528			
	Ebn at Noon	279	299	305	299	294	291	285	279	284	279	272	269			
	Edh at Noon	20	24	28	34	35	35	35	35	29	23	20	18			
	RadAvg	553	869	1223	1578	1858	2124	2219	1853	1420	916	592	440			
(44) All-Sky Solar Radiation	RadStd	38	74	97	91	180	158	90	112	96	95	62	23			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page